

Luke Wiley

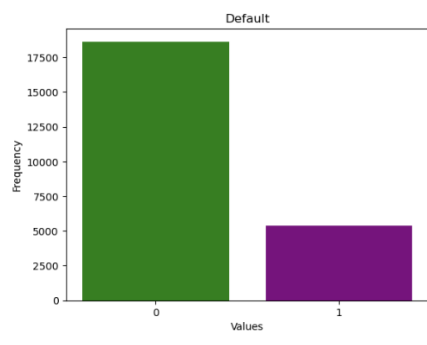
1.a) In order to understand the data set, I spent some time looking at the first 20 objects, where I noticed some odd values, notably, negative bill amounts. From here I decided to check all of the columns for the extraneous values.

- Education: values 0,5,6 outside of range, recode to 4(others).
- Marriage: value 0 outside range, recode to 3(other).
- Bill Amount: Negative values, may be meaningful, no changes.
- Pay Amount: Negative values, may be meaningful, no changes.

Everything else only had valid values.

1.b) To find any missing data, I checked to see if there were any columns where values didn't exist using `data.isna()`. This returned all values false, meaning no missing data. If there were missing values in the data, I could remedy it a couple ways. First I could omit the object from the data. Second, If it were in a column where there is an option for "other" I would recode it to have this value, or had coded it to the mode of the rest of the data to affect the set as little as possible. If it is numeric, I could either code it to "0" or I could make it the average of the rest of the data, depending on what makes more sense in the context.

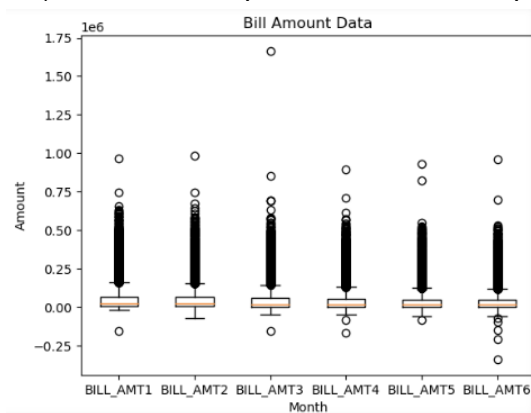
1.c) The target attribute I chose was "DEFAULT", where I found the summary statistics. From this, I was able to gather that the data set was unbalanced, far more non-default cases (18633) than default cases(5367).



2.a) I used `data[bill amounts].describe()` to get the summary statistics of each bill amount column, shown below.

	BILL_AMT1	BILL_AMT2	BILL_AMT3	BILL_AMT4	BILL_AMT5	BILL_AMT6
count	24000.00	24000.00	24000.00	24000.00	24000.00	24000.00
mean	51069.80	49050.72	46870.75	43104.47	40229.08	38779.37
std	73745.22	71162.25	69436.39	63907.82	60576.64	59440.28
min	-154973.00	-69777.00	-157264.00	-170000.00	-81334.00	-339603.00
25%	3490.00	2935.75	2634.50	2315.00	1726.25	1225.75
50%	21986.00	20897.00	19917.00	18937.50	18018.00	16933.50
75%	66822.75	63816.50	59873.50	54444.50	50171.50	49146.25
max	964511.00	983931.00	1664089.00	891586.00	927171.00	961664.00

2.b) I created a plot to show the box plots of each of the 6 bill amount columns, shown below.



2.c) From the data and the plot, we can see that the bill amounts tend to be on the lower end with large outliers which logically makes sense, as most people have to be conservative with their money. The box plots remain consistent over the months, shown as almost identical in the box plots, but a slight downward trend across months from the summary statistics. This would lead one to believe that people were more conservative with their money over time. There are large outliers in the positive direction, representing large bills, but also in the negative direction, maybe representing overpayments.

3.a) Chi-Square tests how independent categorical variables are from one another. We can see the association between two variables by comparing the observed frequencies versus the expected frequencies of how often the two variables appear together.

Mutual Information tests how much we can test about a variable by knowing another. These can be either numeric or categorical, but works best for numeric values, and this is how I used it in the assignment

3.b) To directly apply the Chi-Square function, we need to use it on categorical data, i.e. ordinal or nominal data. This includes: sex, education, marriage, and pay 1 - 6. For Mutual information, we can use it on numeric data, i.e. interval or ratio data. This includes: limit balance, age, bill amount 1 - 6, and pay amount 1 - 6.

3.c) Chi-Square Rank:

cat	feature	chi2 score
0	PAY_1	2696.09
1	PAY_2	2085.12
2	PAY_3	1545.41
3	PAY_4	1359.28
4	PAY_5	1270.26
5	PAY_6	1085.46
6	EDUCATION	49.92
7	SEX	27.09
8	MARRIAGE	9.01

Mutual Info Rank:

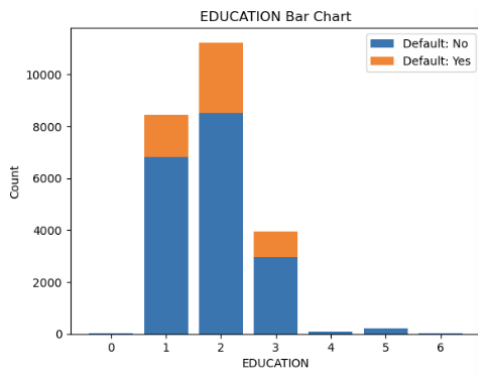
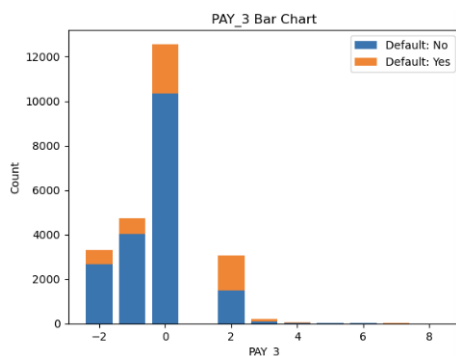
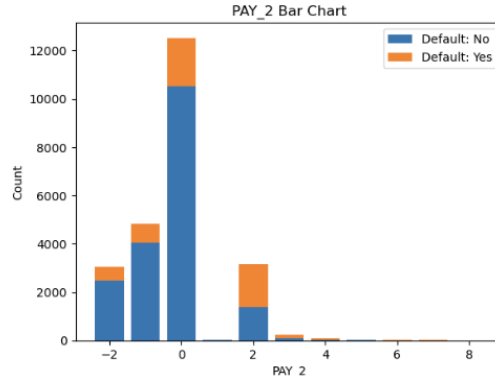
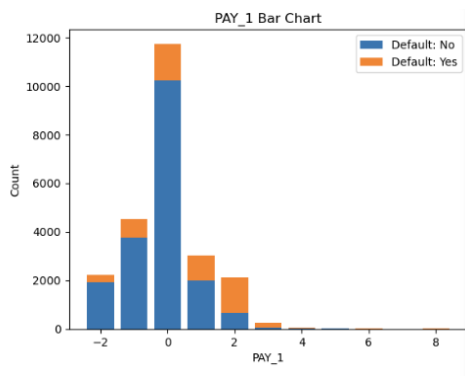
	num	feature	mutual	info
0	PAY_AMT1		0.02	
1	PAY_AMT3		0.02	
2	PAY_AMT4		0.02	
3	PAY_AMT2		0.02	
4	PAY_AMT5		0.01	
5	LIMIT_BAL		0.01	
6	BILL_AMT2		0.01	
7	BILL_AMT1		0.01	
8	PAY_AMT6		0.01	
9	BILL_AMT6		0.01	
10	BILL_AMT5		0.01	
11	BILL_AMT3		0.01	
12	BILL_AMT4		0.00	
13	AGE		0.00	

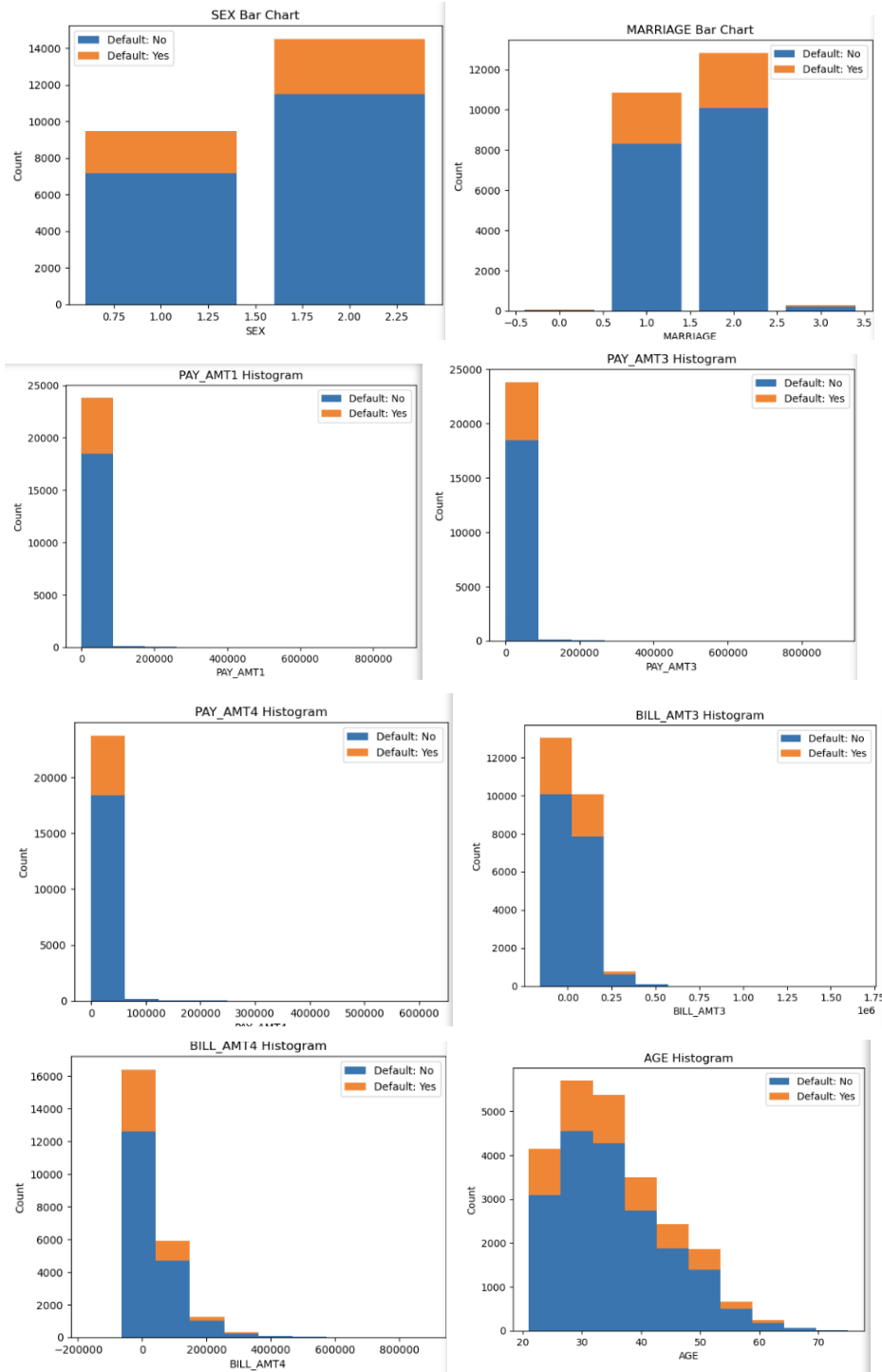
3.d) Top 3 Categorical (PAY_1, PAY_2, PAY_3): We can see a clear trend of the data, where the proportion of default = yes increases as you move right across the graph.

Bottom 3 Categorical (EDUCATION, SEX, MARRIAGE) : The proportion of default = yes to default = no looks quite similar across the graph.

Top 3 Numerical (PAY_AMT1, PAY_AMT3, PAY_AMT4): All of the pay amount are very heavily skewed to the left, with any value other than the ones seen in the larges bar are hard to see, leading me to question the validity of the trend.

Bottom 3 Numerical (BILL_AMT3, BILL_AMT4, AGE): The ratio of default = yes to default = no seems to remain constant across the graph.





4.a) The new feature I generated for data mining is AVG_PAY_TIME, which takes the average pay time from an object over the 6 months. This can be used to describe an object's pay behavior without having to analyse the data itself.

4.b) By normalizing the numeric features of the data set, we can make the features all fall within the same range [0-1]. I did this by applying min-max normalization, using the formula given in lecture. This allows for all features to contribute equally for the modeling process.

4.c) By applying one-hot encoding we can achieve a similar state to normalization but for categorical attributes. It creates a new column for each specific value of a feature that appears and gives its magnitude. By creating 77 new features I was able to get the data in a state where we can perform the modeling process.

5.a) By setting up model 1 on normalized data and model two on unnormalized data (both on one-hot categorical data), we can see the difference normalization had on our results.

5.b) The normalization scheme used was min-max normalization.

Confusion Matrix: four values [True Positive, True Negative, False Positive, False Negative]

True positive: Correctly predicted default

True Negative: Correctly predicted non-default

False Positive: Incorrectly predicted default

False Negative: Incorrectly predicted non-default

Precision: Proportion of predicted defaults that were defaults

Recall: Proportion of defaults correctly identified

F1 - Score: The product of precision and recall divided by their sum multiplied by 2.

Accuracy: Proportion of total predictions correct.

From the report, there is not much difference between the values of the models. This means that normalization did not have a large effect on the modeling process. The values are seen below.

Decision Tree Model 1 Normalized

Precision: 0.3861

Recall: 0.4068

F1: 0.3961

Accuracy: 0.7227

Decision Tree Model 2 Unnormalized

Precision: 0.3865

Recall: 0.4074

F1: 0.3966

Accuracy: 0.7229

6.a) We perform feature selection to reduce the complexity of the model, having it work with fewer total features, and increasing its efficiency, speeding up the process.

6.b) From Task 3, the data set had feature selection performed on it by Chi-Square for categorical attributes, and Mutual Info for numerical attributes. We can then create partial data

sets with only a specific number of features for the modeling process to see how it affects the outcome.

6.c) By performing modeling on these partial sets, we can compare them to the baseline set to see the differences. From the results, we can see that accuracy remains almost identical across sets, but recall seems to decrease with less features. Precision remains similar, except for the Top 1 categorical + Top 1 Numerical, being larger than the rest. The F1 score seems to vary across sets. None of the value changes were very significant, so we can conclude that feature selection simplified the model while not hurting performance to a noticeable degree. The values can be seen below.

```
Baseline
Precision : 0.3905
Recall    : 0.4196
F1        : 0.4045
Accuracy  : 0.7238

Top 1 categorical, Top 1 numerical
Categorical: ['PAY_1'] | Numerical: ['PAY_AMT1']
Precision : 0.5089
Recall    : 0.3223
F1        : 0.3946
Accuracy  : 0.7788

Top 3 categorical, Top 3 numerical
Categorical: ['PAY_1', 'PAY_2', 'PAY_3'] | Numerical: ['PAY_AMT1', 'PAY_AMT3', 'PAY_AMT4']
Precision : 0.3916
Recall    : 0.3589
F1        : 0.3745
Accuracy  : 0.7320

Top 5 categorical, Top 5 numerical
Categorical: ['PAY_1', 'PAY_2', 'PAY_3', 'PAY_4', 'PAY_5'] | Numerical: ['PAY_AMT1', 'PAY_AMT3', 'PAY_AMT4', 'PAY_AMT2', 'PAY_AMT5']
Precision : 0.3860
Recall    : 0.3810
F1        : 0.3833
Accuracy  : 0.7260
```